

ZIELONA GORA, Poland

WMO: 124000

Lat: 51.9299N

Lon: 15.5247E

Elev: 631

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	14.7	3.8	6.9	11.6	9.2	9.0	18.0	20.9	42.9	18.9	41.9	6.1	70	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	86.2	66.6	82.7	65.3	79.5	64.2	69.2	80.9	67.5	78.1	65.9	75.9	6.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	96.2	72.4	63.9	91.0	71.0	62.4	86.2	70.1	33.6	81.5	32.3	77.7	31.0	76.2	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	14.4	12.8	DB	5.4	92.2	7.3	2.9	0.2	94.3	-4.1	96.0	-8.3	97.7	-13.6	99.8	
			WB	4.8	72.2	7.2	1.8	-0.4	73.5	-4.6	74.6	-8.6	75.5	-13.8	76.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	30.8	33.3	39.3	49.2	57.2	63.4	67.3	66.6	58.3	49.2	40.0	33.4
	DBStd	14.91	9.15	8.61	7.87	8.10	7.18	6.72	6.37	5.96	6.03	7.19	7.07	8.10
	HDD50	2462	596	468	342	111	16	0	0	0	4	104	305	516
	HDD65	6126	1061	887	798	477	258	110	48	52	215	490	750	980
	CDD50	2127	0	1	8	86	241	401	537	514	254	80	5	0
	CDD65	318	0	0	0	3	18	61	120	101	15	0	0	0
	CDH74	2412	0	0	0	26	147	479	951	709	99	1	0	0
	CDH80	640	0	0	0	2	22	127	289	184	16	0	0	0
Wind	WSAvg	6.7	7.5	7.5	7.4	6.9	6.3	6.2	6.1	5.8	5.9	6.4	6.7	7.6
Precipitation	PrecAvg	23.7	1.7	1.4	1.8	1.3	2.1	2.4	3.5	2.6	1.9	1.5	1.7	1.6
	PrecMax	30.4	4.1	3.4	5.0	2.8	5.0	4.5	8.2	5.2	5.6	3.9	4.7	3.9
	PrecMin	15.6	0.1	0.2	0.7	0.1	0.4	0.5	0.8	0.7	0.2	0.4	0.0	0.5
	PrecStd	4.2	0.9	0.7	1.0	0.7	1.1	1.1	2.0	1.3	1.1	0.9	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.5	55.6	64.4	77.5	83.0	88.8	91.1	90.4	82.3	71.7	58.7	52.7
		MCWB	47.5	47.9	50.9	59.3	63.6	68.2	67.4	68.0	65.3	61.2	53.4	48.8
	2%	DB	48.4	50.9	58.4	71.9	78.3	83.6	86.7	84.7	76.6	66.5	55.1	49.4
		MCWB	45.4	45.7	48.3	56.4	61.9	65.7	66.3	66.7	62.3	58.3	50.7	46.3
	5%	DB	45.3	47.9	54.2	66.6	74.1	79.4	83.0	80.8	72.6	62.8	52.2	46.6
		MCWB	42.3	43.4	46.3	53.5	59.8	63.4	65.4	65.3	61.0	56.4	48.9	44.1
	10%	DB	42.5	45.0	50.6	62.7	70.2	75.6	79.3	77.7	68.7	59.4	49.8	44.0
		MCWB	40.1	41.4	44.5	51.5	57.6	61.9	64.4	64.2	59.4	54.8	47.2	41.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	49.4	52.8	61.1	65.7	71.0	70.8	70.8	66.9	62.2	54.6	49.9
		MCDB	50.7	52.9	61.2	74.2	78.1	84.5	82.9	83.8	77.4	69.9	57.8	51.8
	2%	WB	45.6	46.5	49.8	57.4	63.4	67.7	69.1	68.9	64.4	59.4	51.4	46.8
		MCDB	48.2	49.9	56.2	68.8	74.9	79.0	80.6	80.3	73.1	64.9	53.9	48.9
	5%	WB	42.7	44.0	47.5	54.7	61.3	65.4	67.4	67.3	62.4	57.1	49.5	44.3
		MCDB	44.9	47.3	52.7	64.5	71.4	75.9	78.3	77.1	70.0	61.2	51.9	46.2
	10%	WB	40.3	41.5	45.3	52.5	59.2	63.3	65.8	65.6	60.6	55.2	47.3	42.0
		MCDB	42.3	44.8	49.8	61.1	67.6	72.5	75.9	74.5	67.1	58.9	49.5	43.8
Mean Daily Temperature Range		MDBR	7.9	9.5	11.9	15.5	16.3	16.0	15.7	15.9	14.1	11.6	8.6	7.4
	5% DB	MCDBR	10.3	12.7	17.9	22.1	22.3	22.0	21.0	21.0	20.4	15.9	12.1	9.7
		MCWBR	8.2	8.8	10.3	10.8	10.2	9.4	7.5	8.7	9.5	9.2	8.6	7.9
	5% WB	MCDBR	10.4	11.5	15.7	19.9	19.6	20.1	18.5	18.6	17.9	14.6	11.8	9.3
		MCWBR	8.8	8.9	10.3	10.2	9.8	9.7	7.8	8.6	9.6	9.0	9.0	7.9
Clear-Sky Solar Irradiance	taub	0.312	0.336	0.373	0.413	0.409	0.407	0.427	0.422	0.382	0.360	0.338	0.306	
	taud	2.355	2.341	2.263	2.209	2.266	2.299	2.251	2.281	2.366	2.395	2.393	2.372	
	Ebn at Noon	218	246	257	260	268	269	261	255	253	234	206	200	
	Edh at Noon	20	26	34	41	40	39	41	37	31	25	19	16	
All-Sky Solar Radiation	RadAvg	222	440	775	1286	1566	1662	1572	1374	991	559	255	161	
	RadStd	32	80	102	179	182	132	194	130	130	75	40	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.95	+1.08	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.72	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	+94	N/A

Nomenclature: See separate page